Development of a Convolutional Neural Network-Based Chrysanthemum Variety Recognition System

Project Overview:

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- **Project Type:** Individual undergraduate dissertation
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Abstract:

This project developed an image recognition system using convolutional neural networks and transfer learning. An EfficientNet-B3 model was developed using a dataset of 1,169 images across eight classes, with separate training, validation, and test splits. The trained model was integrated into a Flask web application. The interface supports image upload, preview, and the display of the three highest-ranked predictions with confidence scores. The saved model achieved 95.38% validation accuracy.

Introduction:

Similar flower appearances and variation in lighting, pose, and background can make visual identification difficult. The project explored how a pretrained image classifier could be adapted to a small flower dataset and made accessible through a practical web interface.

Key Contributions:

Dataset Preparation:

- Collected and organised images into eight classes, checked for damaged files, and prepared the data for model development.

Model Development:

- Adapted a pretrained EfficientNet-B3 network with a custom eight-class classifier and selectively fine-tuned the later feature extraction layers.

Application Development:

- Integrated the trained model with a Flask prediction API and a browser interface for uploading images and viewing ranked predictions.

Methodology:

Data Processing and Augmentation:

- Applied random resized cropping, horizontal flipping, rotation, and colour jitter during training to introduce variation in the input images.
- Used 224 × 224 model inputs and ImageNet normalisation to match the pretrained network preprocessing pipeline.

Transfer Learning and Training:

- Replaced the original classifier with fully connected layers, ReLU activations, and dropout, ending in eight output classes.
- Fine-tuned the final five feature modules and trained the classifier using cross-entropy loss and AdamW, with separate learning rates for the backbone and classifier.
- Used cosine learning-rate scheduling and early stopping, retaining the checkpoint with the best validation accuracy.

Web Application:

- Built a Flask backend that accepts an uploaded image, applies preprocessing, and returns the top three class predictions with confidence scores.
- Created an interface for image selection, preview, and prediction display, connecting model inference to an accessible user workflow.

Results and Impact:

The best saved checkpoint recorded 95.38% validation accuracy. The project produced a working prototype connecting dataset preparation, model training, and web-based inference. Its main outcome was an end-to-end implementation demonstrating how transfer learning can support a focused image classification task.

The result applies to the validation split used during development. Confidence scores represent model predictions and should not be interpreted as a guarantee of correct identification. Performance on unfamiliar images and new acquisition conditions requires further evaluation.

Future Directions:

- Expand and rebalance the dataset with more varied lighting, backgrounds, and viewpoints.
- Extend evaluation to an additional external dataset and report reproducible per-class precision, recall, and confusion patterns.
- Explore rejection of unknown classes and optimise inference speed for practical use.